Experiment – 6

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Semester: 5th
Date of Performance: 10/8/26
Subject Name: ADBMS
Subject Code: 24CSP-310

(A)

A fruit distribution company records the number of apples and oranges sold each day. For every date, exactly one record exists for apples and one record exists for oranges.

Generate a daily sales comparison report showing the difference between the number of apples sold and the number of oranges sold on the same day. The difference should be calculated as:

Difference = Apples Sold - Oranges Sold

Display the sale date and the calculated difference. The result should be sorted by sale date in ascending order.

Sales

sale_date	fruit	sold_num
2020-05-01	apples	10
2020-05-01	oranges	8
2020-05-02	apples	15
2020-05-02	oranges	15
2020-05-03	apples	20
2020-05-03	oranges	0
2020-05-04	apples	15
2020-05-04	oranges	16

Output

sale_date	diff
2020-05-01	2
2020-05-02	0
2020-05-03	20
2020-05-04	-1

Solution:

```
CREATE TABLE Sales
(
    sale_date DATE,
    fruit VARCHAR(20),
    sold_num INT,
    PRIMARY KEY (sale_date, fruit)
);

INSERT INTO Sales
VALUES
('2020-05-01','apples',10),
```

```
('2020-05-01','oranges',8),

('2020-05-02','apples',15),
('2020-05-02','oranges',15),

('2020-05-03','apples',20),
('2020-05-03','oranges',0),

('2020-05-04','apples',15),
('2020-05-04','oranges',16);


SELECT sale_date,
    SUM(
        CASE
            WHEN fruit = 'apples' THEN sold_num
            ELSE -sold_num
        END
    ) AS diff
FROM Sales
GROUP BY sale_date
ORDER BY sale_date;
```

(B)

A company stores hierarchical employee reporting information in a table. Each record represents a node in a hierarchy where:

- id represents the current node.
- p_id represents the parent node.

Classify every node into one of the following categories:

- **Root**: A node that does not have any parent.
- **Leaf**: A node that has a parent but does not have any children.
- **Inner**: A node that has both a parent and at least one child.

Display the node ID along with its category. Return the result sorted by node ID.

Tree

id	p_id
1	NULL
2	1
3	1
4	2
5	2

Expected Output

id	Type
1	Root
2	Inner
3	Leaf
4	Leaf
5	Leaf

Solution:

```
CREATE TABLE Tree
(
    id INT PRIMARY KEY,
    p_id INT
);

INSERT INTO Tree
VALUES
(1,NULL),
(2,1),
(3,1),
(4,2),
(5,2);

SELECT ID,
      CASE
            WHEN P_ID IS NULL THEN 'ROOT'
            WHEN ID NOT IN (
                  SELECT P_ID
                  FROM TREE
                  WHERE P_ID IS NOT NULL
            ) THEN 'LEAF'
            ELSE 'INNER'
      END AS "TYPE"
FROM TREE;
```